

You Only Look At Once
(YOLO)

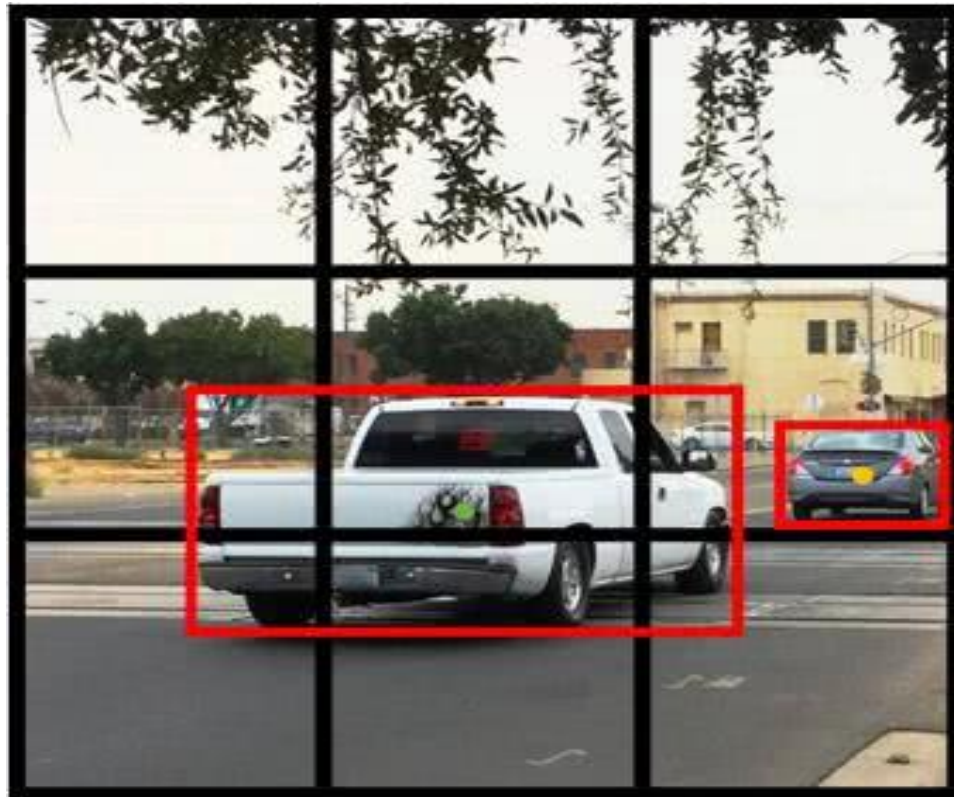
YOLO Idea

- Detection problem requires variable number of outputs. It is not possible to design a NN with variable number of outputs.
- YOLO solves the issue by associating variable objects at cell level, thus making it fixed number of outputs.



How do we associate objects to cells?

YOLO algorithm divides a training image into grids and assigning an object to a grid if and only if the center of the object falls inside the grid.

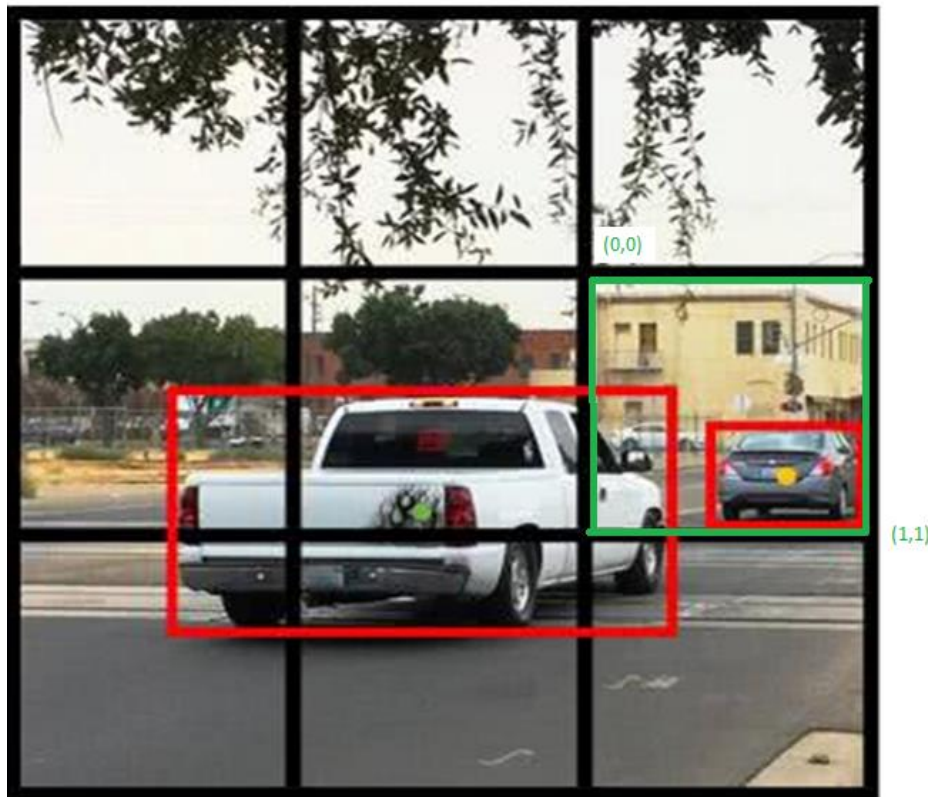


Assume that there are 3 types of targets we're detecting:

- pedestrian
- car
- motorcycle

How do we associate objects to cells?

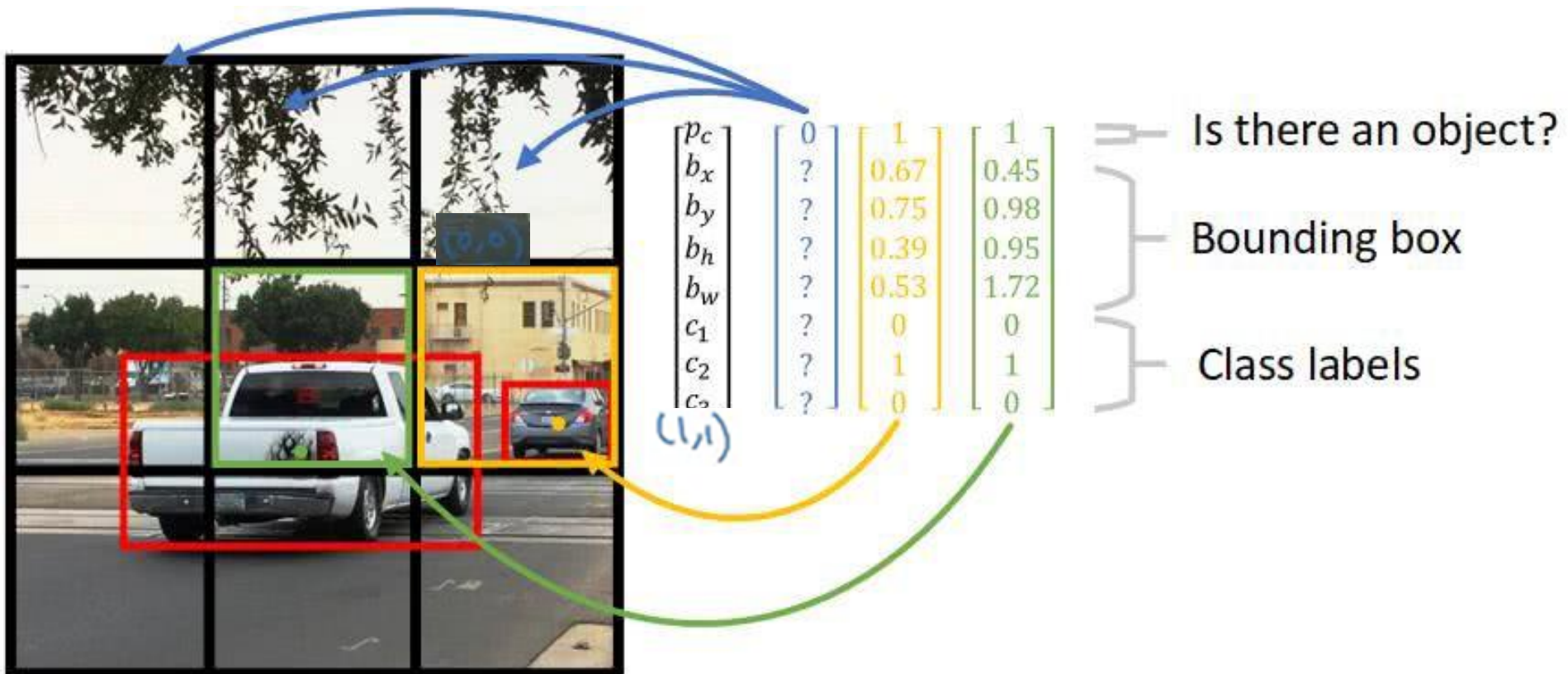
Each object in a training image can get assigned to exactly one grid and then the corresponding bounding box is represented by the coordinates relative to the grid.



$$y = \begin{bmatrix} 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \left. \begin{array}{l} \text{between } 0 \text{ and } 1 \\ \text{could be } > 1 \end{array} \right\}$$

$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

How do we associate objects to cells?

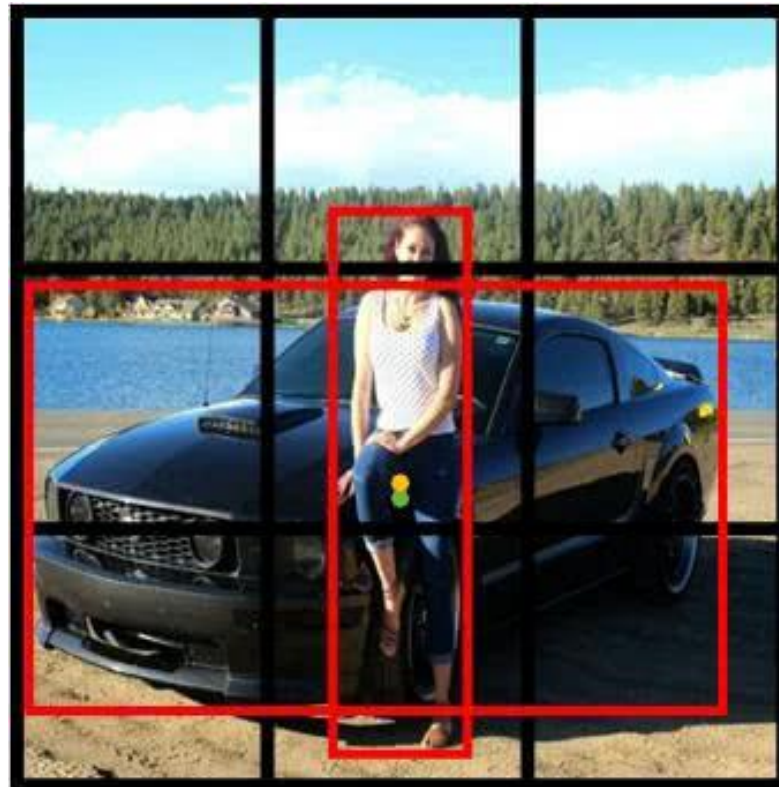


How do we associate objects to cells?

- For the rightmost car, it's easy. It belongs to the middle right cell since its bounding box is inside that grid cell.
- For the truck in the middle of the image, its bounding box intersects with several grid cells. The YOLO algorithm takes the middle point of the bounding box and associates it to the grid cell containing it.

Multiple objects in same grid cell

For instance, there's a person standing in front of a car and their bounding box centers are so close. Shall we choose the person or the car?



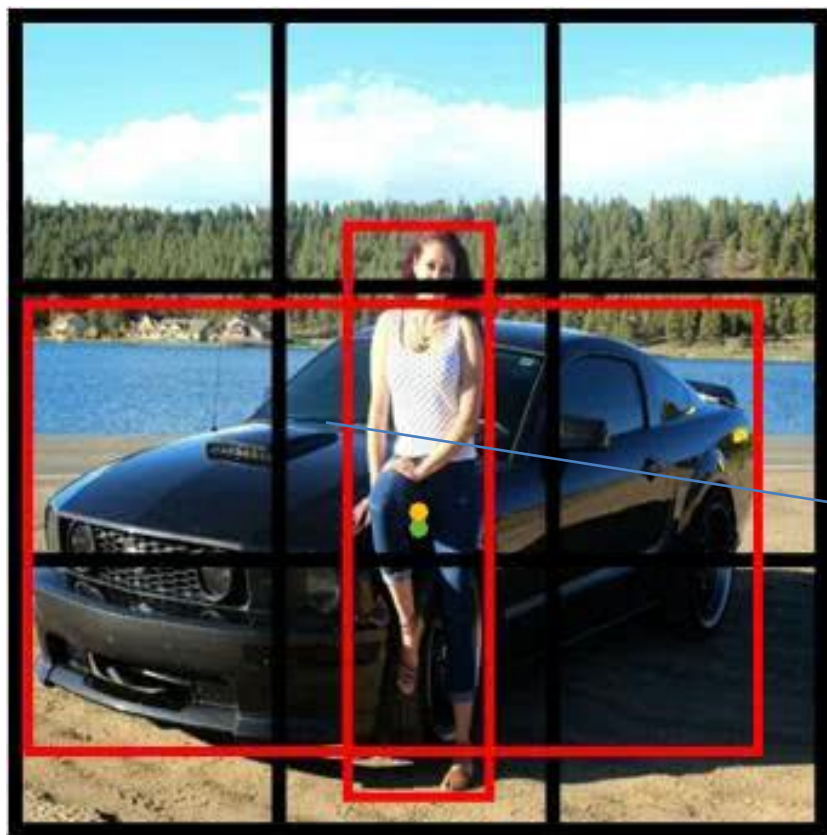
Multiple objects in same grid cell

- Multiple *anchor boxes* (of different shapes) are used to resolve the problem.
- Each grid cell now has two anchor boxes, where each anchor box acts like a container. Meaning now each grid cell can predict up to 2 objects.
- As a result, the output of one grid cell will be extended to contain information for two anchor boxes.

Why anchor boxes with different shapes?

- Place the object in the anchor box with which object bounding box has higher IOU score.
- For the above example, the person will be associated with the tall anchor box since their shape is more similar.

Multiple objects in same grid cell

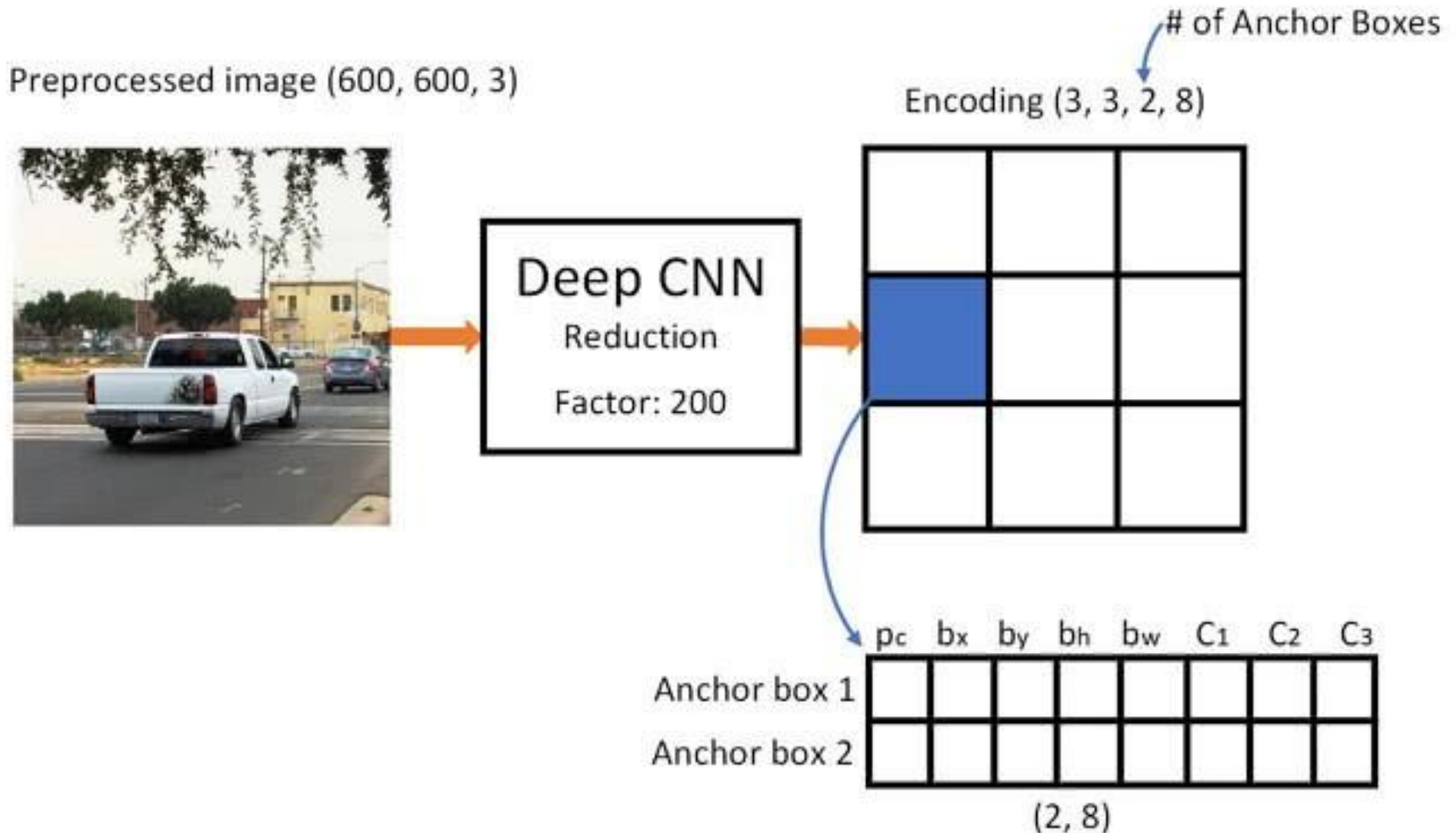


$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \\ p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 1 \\ 0 \\ 0 \\ 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

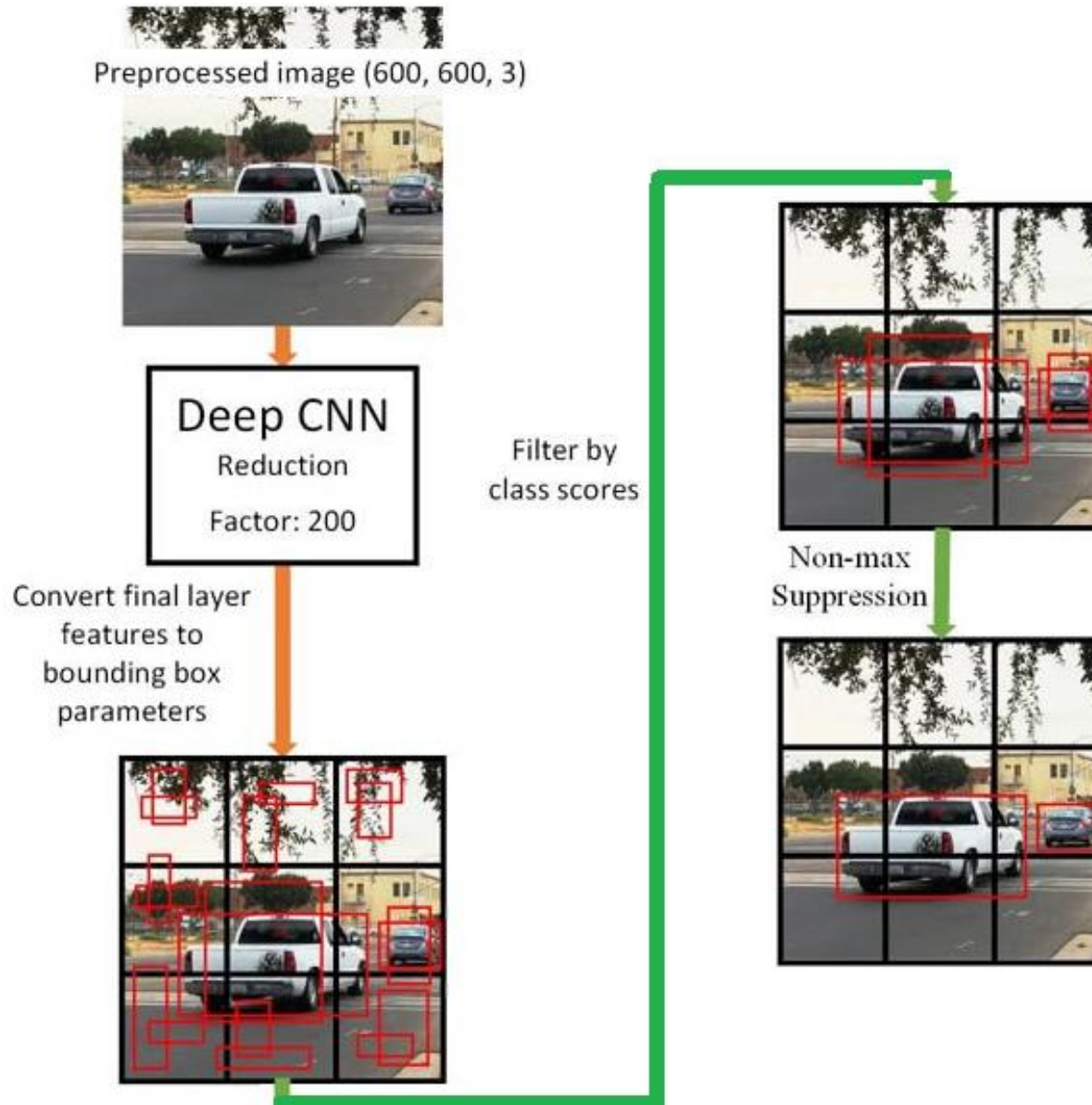
Anchor box 1
Pedestrian

Anchor box 2
Car

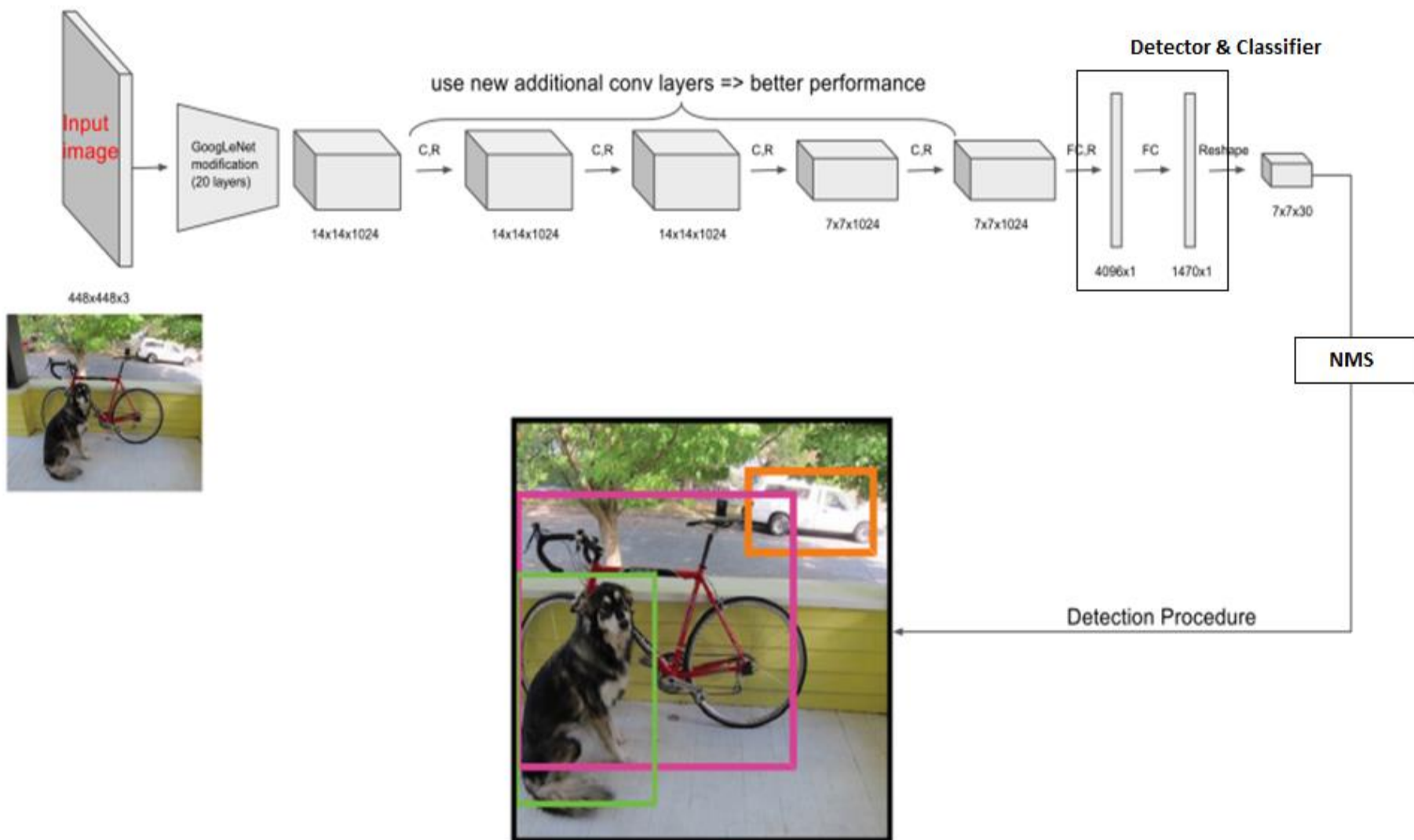
YOLO Training



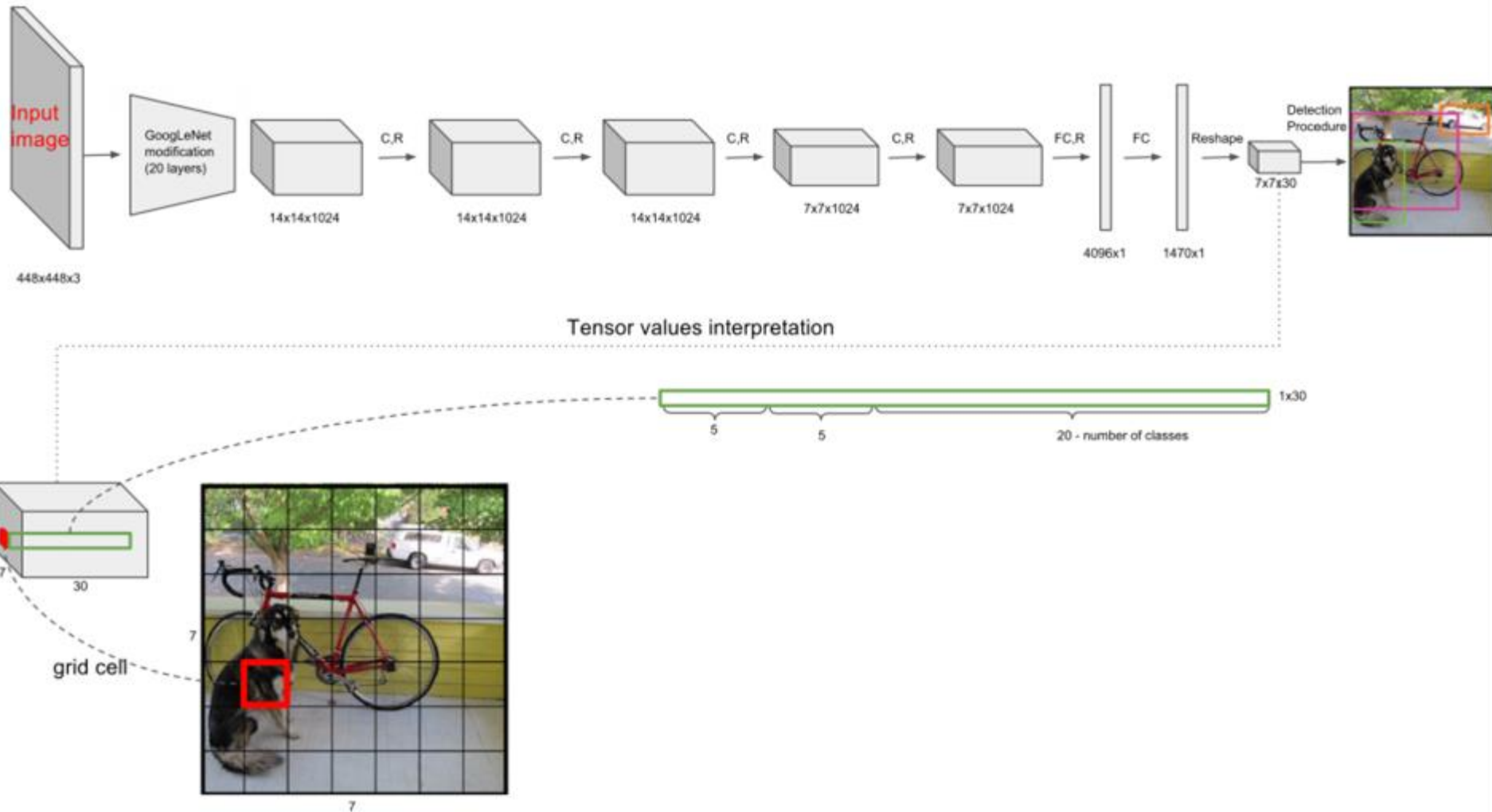
YOLO Prediction



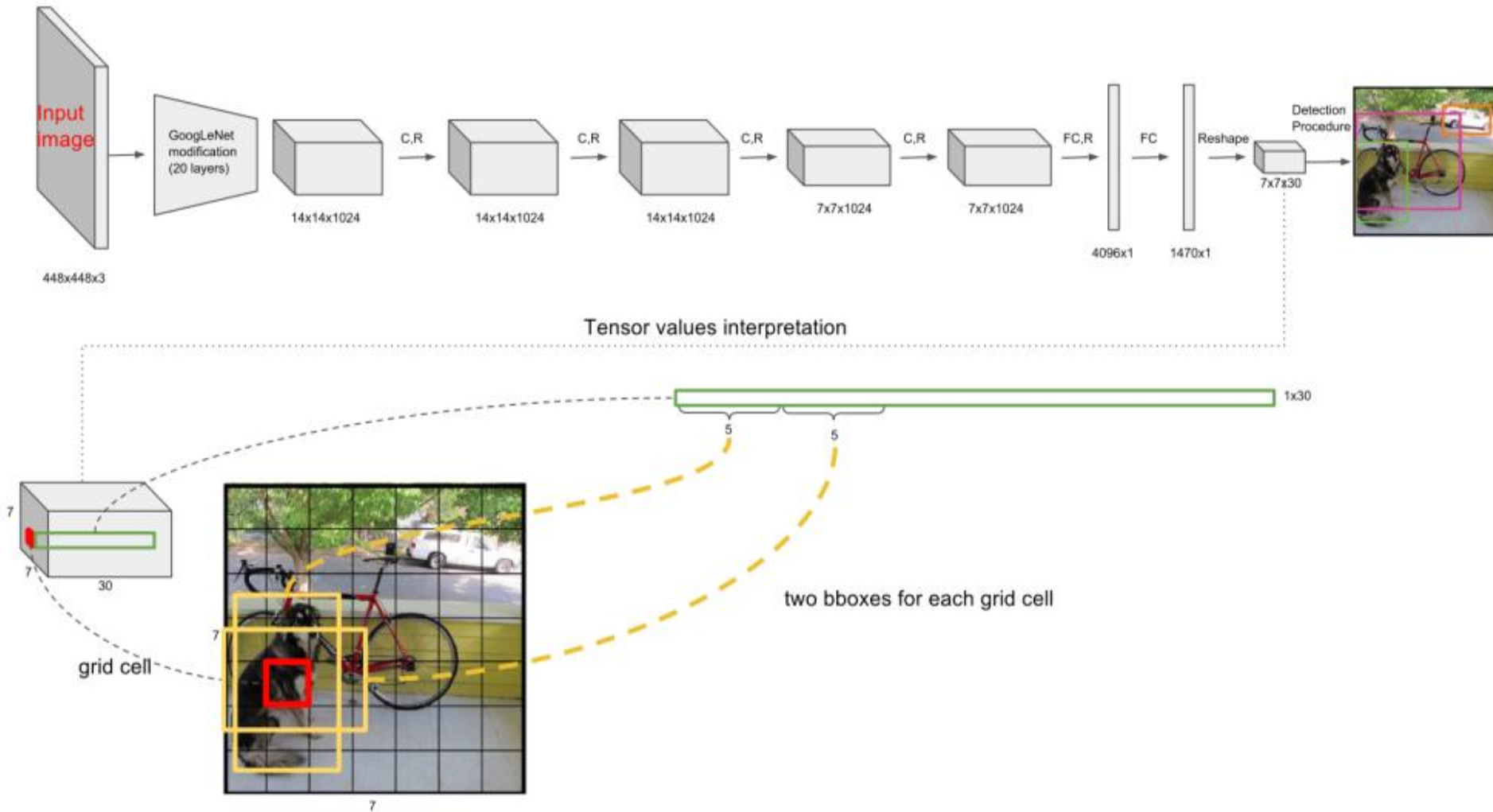
Model Architecture



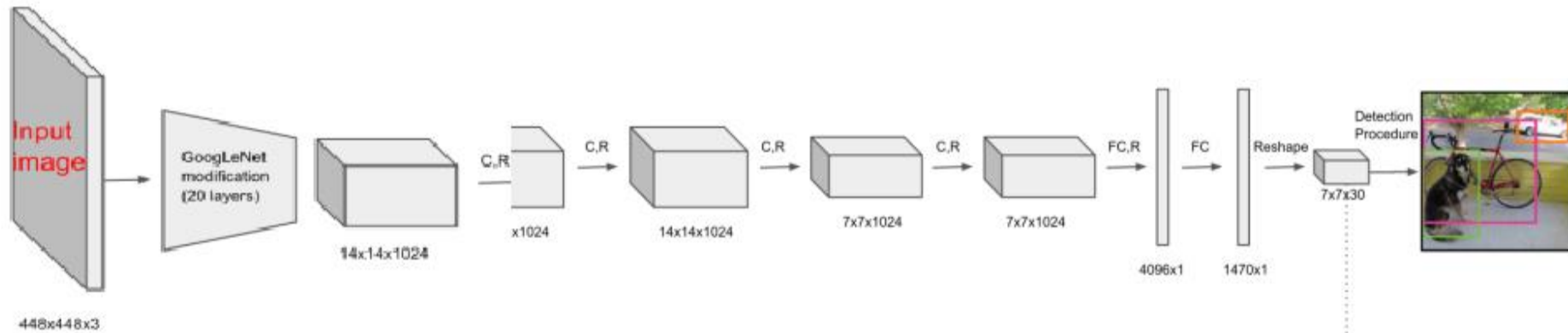
YOLO Training



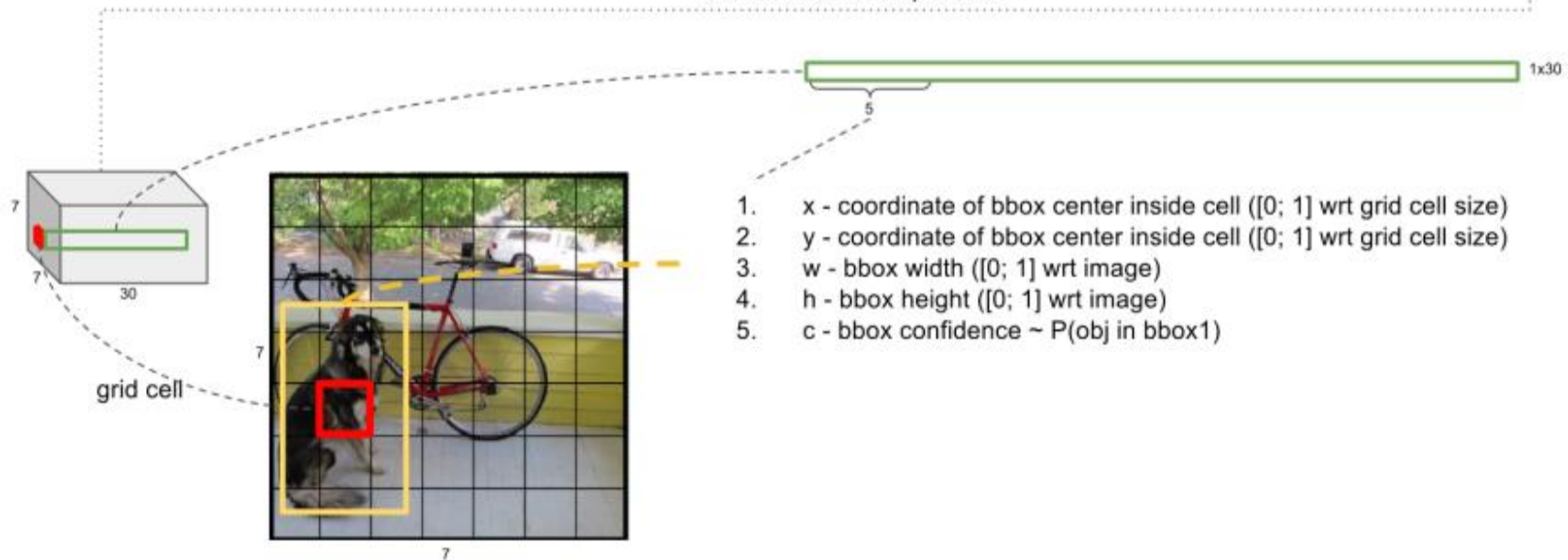
YOLO Training



YOLO Training



Tensor values interpretation



YOLO Training: Loss function

Note that the loss function only penalizes classification error if an object is present in that grid cell (hence the conditional class probability discussed earlier). It also only penalizes bounding box coordinate error if that predictor is “responsible” for the ground truth box (i.e. has the highest IOU of any predictor in that grid cell).

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2$$

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2$$

$$+ \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2$$

$$\mathbb{1}_{ij}^{\text{obj}}$$

The j th bbox predictor in cell i is “responsible” for that prediction

$$\mathbb{1}_{ij}^{\text{noobj}}$$

$$\mathbb{1}_i^{\text{obj}}$$

If object appears in cell i

YOLO Training: Loss function

$$\begin{aligned}
 & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\
 & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\
 & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\
 & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\
 & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \quad (3)
 \end{aligned}$$

model. We use sum-squared error because it is easy to optimize, however it does not perfectly align with our goal of maximizing average precision. It weights localization error equally with classification error which may not be ideal. Also, in every image many grid cells do not contain any object. This pushes the “confidence” scores of those cells towards zero, often overpowering the gradient from cells that do contain objects. This can lead to model instability, causing training to diverge early on.

To remedy this, we increase the loss from bounding box coordinate predictions and decrease the loss from confidence predictions for boxes that don't contain objects. We use two parameters, λ_{coord} and λ_{noobj} to accomplish this. We set $\lambda_{\text{coord}} = 5$ and $\lambda_{\text{noobj}} = .5$.

$$\lambda_{\text{coord}}=5, \lambda_{\text{noobj}}=0.5$$

YOLO Training: Loss function

Sum-squared error also equally weights errors in large boxes and small boxes. Our error metric should reflect that small deviations in large boxes matter less than in small boxes. To partially address this we predict the square root of the bounding box width and height instead of the width and height directly.

$$\begin{aligned} & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\ & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\ & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\ & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\ & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \end{aligned}$$

YOLO Training: Loss function

epochs=135

batch_size=64

momentum_a = 0.9

decay=0.0005

lr=[10^{-3} , 10^{-2} , 10^{-3} , 10^{-4}]

dropout_rate=0.5

augmentation

=[scaling, translation, exposure, saturation]

Limitation of YOLO

- Group of small objects
- Unusual aspect ratios
- Coarse feature
- Localization error of bounding box